

Innovation in Teacher Education: The Evolution of a Program

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Successful innovation in teacher education depends upon resolving issues regarding four interrelated themes: identity, non-passivity, control, and curriculum. The authors' examination of the first year and a half of an innovative masters-level program explores the specific development of these themes. For example, faculty and prospective teachers began to establish an identity with teaching and teacher education rather than with subject specializations. McCaleb and his colleagues discuss their institutional progress on the radical changes needed in teacher education.

The preparation of teachers is an endangered role of colleges and universities. Critics from within and from without have pronounced serious indictments (e.g., Conant, 1963; Feistritzer, 1984; Koerner, 1963; Lyons, 1979; Weaver, 1983). Despite these attacks, teacher education appears unable to defend itself and incapable of correcting the problems. In 1970 Silberman charged "the preparation of teachers has been studied as frequently as the plight of the black man in America, and with as little effect" (p. 414), and 16 years later "the troublesome circumstances remain basically unchanged" (Lanier and Little, 1986, p. 527).

Giving up on schools of education, some critics call for the elimination of traditional university-based teacher education programs. Proposed alternatives range from simply disposing of teacher preparation and certifying liberal arts graduates (Southern Regional Education Board, 1984; Toch, 1983), to transferring teacher education from universities to public school systems (Schlechty, 1985), to extending teacher preparation into a post-baccalaureate program and eliminating undergraduate teacher preparation programs (Carnegie Forum's Task Force, 1986; Holmes Group, 1986). What is necessary for schools of education to mount an effective innovation?

Committed to the belief that innovation in teacher education is both necessary and possible, and that the university's role in designing, implementing, and evaluating innovation is crucial, the University of Maryland initiated a pilot program in 1985. Now in the second cycle of this program, we believe that the innovation is effective. Successful innovation requires significant curricular reform, and we conclude that such reform depends upon

resolving issues in *passivity, identity, and control*.

Reforming Teacher Education Curriculum

An obvious target for innovation is the curriculum. "The research is unequivocal about the general, overall course work provided for teachers. It remains casual at best and affords a poorly conceived collage of courses across the spectrum of initial preparation" (Lanier and Little, 1986, p. 549). Courses are criticized as being too shallow and too technical (e.g., Beyer and Zeichner, 1982; Carnegie Forum's Task Force, 1986; Conant, 1963).

Increasing the rigor of the curriculum may be impeded by characteristics of teacher educators and prospective teachers. Teacher educators' intellectual performance is sometimes characterized by "a lack of probing thought and analysis" (Lanier and Little, 1986, p. 535). Prospective teachers' below-average academic abilities "depress the levels of content knowledge assumed appropriate for the teacher education curriculum and discourage talented teachers and students from being seriously committed to the improvement of teaching and teacher education" (Lanier and Little, 1986, p. 541).

Attempts at reform have failed by attending to the surface-level problems. Courses have been redesigned, education classes have been eliminated, and fifth years have been added. However, underlying, foundational issues have been untouched, perhaps undetected. In order to make lasting innovations in the curriculum, the deeper themes of passivity, identity, and control must be addressed.

Passivity

A recurring theme in critiques of

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teacher education is its passivity. Teacher education is cast as the pre-liberation female, the acted-upon rather than the actor or agent. Teacher education not only lacks a sense of efficacy, her consciousness has not yet been raised sufficiently to recognize or become angry about her subordination. Teacher education has not taken a proactive stance, a position of the creator, but has "evolved as a creature of teaching" (Holmes Group, 1986, p. 26). While many conditions of teaching could be listed, critical ones forcing a passive role include the lack of autonomy and the limitation of reflective problem solving (Denemark, 1985; Howsam, Corrigan, Denemark, and Nash, 1976).

Berliner (1984) captures the plight of teacher education in its timidity, ignorance, and lack of vision: teacher educators accept "mediocrity in teacher education" and shy away from "reading, critiquing, or using the scientific literature about teaching" (pp. 1-2). Lanier and Little (1986) echo this concern with their conclusion about the faculty in teacher education who "assume professional work assignments and routines that demand minimal intellectual flexibility and breadth and require, instead, conformity and limited analysis" (p. 535). Change in anything, including curriculum, does not come from conformists. Unless this characteristic passivity of teacher educators is replaced with a more active orientation, innovations in teacher education are destined to fail.

One way of casting an active orientation is to conceptualize the teacher as a reflective, problem-solving professional. The Carnegie Forum (1986) recommends taking "maximum advantage of the research on teaching and the accumulated knowledge of exceptional teachers" and cultivating the habit of reflecting on teaching (p. 50). Unfortunately, many prospective teachers seem to adopt the same degree of comfort with the status quo as those who educate them. For example, graduates of teacher education programs overwhelmingly select student teaching as their most important experience, a setting where "the emphasis is upon imitation of the subservience to the supervising teacher, not upon investigation, reflection, and solving

novel problems" (Holmes Group, 1986, p. 62). Berliner (1984) also concludes that student teaching in its traditional design "retards the development of analytic skills" (p. 3).

Ironically, although research on teaching engages the active inquiry that has been recommended, the normal procedure of imposing this research into the curriculum reinforces the passive role of the teacher. Teachers are recipients of the interpreted investigations of the researchers. The relationship between the research on teaching and teachers must be reconstructed to promote the active role of the teacher. A possible way to make this change is found in Fenstermacher's (1986) practical arguments where teachers actively apply research findings to their decisions about teaching. Teachers construct the arguments for their actions often drawing their premises from research findings.

Identity

In addition to promoting passive traits, teacher education programs fail to develop a sense of professional identity among teachers and teacher educators: "the curriculum's easy access and implicit assurances of success provide the opportunity to learn that 'anybody can teach'" (Lanier and Little 1986, p. 549). Pedagogy lacks the status and rigor of a traditional discipline. Unable to identify with a strong curriculum of their own, teachers and teacher educators become stepchildren of the content fields or descendants from psychology, sociology, or other more respected disciplines. As members of the Holmes group (1986) suggest:

Reforming the education of teachers depends upon engaging in the complex work of identifying the knowledge base for competent teaching, and developing the content and strategies whereby it is imparted. Although specialized professional knowledge has been under development for some time — and dramatic strides have been made during the past two decades — an amalgam of intuition, unreflective reactions, and personal dispositions still seem to ground the right to teach. Improving teaching requires teachers to act on legitimate professional knowledge, skills, and an ethos of responsibility. (p. 55)

Educating a Profession also articulately portrays the lack of disciplinary identity:

Basic to the problem of developing a professional culture is the long held view that there is not much to teaching, and that most people with more knowledge than the learner can teach. Teaching has been based more upon conventional wisdom, folkways, and personal experience than upon solidly validated professional knowledge and skill. The tradition dies hard. . . . *To fail to develop principles, concepts, and theories, and to validate practice is to restrict the occupation to the level of a craft.* (Howsam, Corrigan, Denemark, and Nash, 1976, p. 11)

Just as identity in teacher education programs has not been built through attraction to a strong and respected discipline, neither has it been developed through structures and events that bind persons together. Lortie (1975) contrasts other occupations with teaching in the formation of identity through shared ordeals. He emphasizes the extent to which teachers work as individuals. Teacher education shows its inheritance as the creature of teaching — prospective teachers and teacher educators typically work in isolation, thereby impeding any feeling of group identity. Even when the curriculum presents an ordeal such as student teaching, prospective teachers seldom experience the ordeal as a cohort and thus have little opportunity to build group identity.

Neither prospective teachers nor teacher educators embrace their ties to teacher education. Lanier and Little (1986) show how extensively the faculty in colleges of education avoid a primary identification as teacher educators. Similarly, prospective teachers resist identifying with their preparation program and instead bring expectations which "stand in contrast to those of persons in other professions who anticipate difficult access, hard work, a sense of academic value, and occupational continuation" (Lanier and Little, 1986, p. 543). Given the American value for hard work, persons prefer to identify with a rigorous program, a reputation not generally held by the teacher education program.

Control

Clearly, innovation in teacher education depends upon making particular changes in the curriculum which will alter the norms of passivity and non-identity. However, promising curricular innovations often fail due to the

process of change. Hall (1978) outlines strategies for implementing a change in teacher education, showing none to be failsafe. One of these strategies, the experimental program, appears closest to our process of innovation. This strategy, according to Hall, carries the potential advantage of never becoming institutionalized, particularly because of problems of ownership: "The administration and program staff need to make certain that they are constantly reaching out to the regular faculty members to involve them in the development and evolution of the experimental program" (p. 60).

Crucial to any change process is the selection of decision makers. Who exercises authority? Joyce, Hersh, and McKibbin (1983) elaborate a concept of "responsible parties" as a way of incorporating the essential groups into sharing decisions and responsibility for improvements. Formation of responsible parties means collaborative planning, action, and follow-through by representatives of the significant interest groups. Lanier and Little (1986) assert that the major inhibitor to change and improvement in teacher education is the diffusion of responsibility: "Unless some administrative unit is given the authority and concomitant accountability, teacher education will not be well taken care of" (pp. 557-558). Timid teacher educators are not innovators. Until teacher educators and teachers participate significantly in the control of the process, the thorns of ownership, passivity, and identity leave no growing room for any seeds of change.

In this brief review of the themes of innovation, the curriculum of teacher education appears to be the primary target for reform. However, innovation in curriculum seems unlikely to succeed unless the deeper problems of passivity and identity receive attention. Another hidden obstacle confronting efforts to innovate comes in the process of change. Good ideas are not enough; the way they are planted determines their growth. The authority and accountability for the innovation should be in the hands of teacher educators; and within the group of teacher educators, both those who advocate change and those who represent existing interests must be involved

in the innovation.

The next section presents a description of our program. An abstract discussion of innovation often fails to provide the subtle and complex interactions found in the concrete application. The program description reveals how intertwined curriculum, passivity, identity, and control become in practice. The description generally follows the development of the program over time with attention given to its inception, the participants, the curriculum, and the process of innovation.

Description of the Experimental Program

Inception

The origin of a program often foreshadows the emerging lines of authority and accountability, critical features of control of the program. In the summer of 1985 the faculty of the Department of Curriculum and Instruction at the University of Maryland initiated a postbaccalaureate program that would provide its students with certification to teach. Three earlier actions by the chair of the department had significant implications for the emergence of the program: (a) the appointment of a task force in the fall of 1984 to explore alternatives to the existing undergraduate teacher preparation programs; (b) the encouragement to a few faculty interested in implementing an innovative program and primarily engaged in conducting research in teaching and learning, but who were only marginally involved in the undergraduate teacher preparation program; and (c) involvement by the Maryland State Department of Education to support the innovation with a small grant and to certify its graduates.

A task force of eight people did the earliest structured work: The task force included six faculty representing a cross section of the department, one coordinator of field experiences employed by both a public school system and the university, and a director of staff development for another public school system. The appointment of school-based persons was initially controversial to some departmental faculty. However, because of its diverse and broadly-based representation, the task force had the potential for promoting

the adoption of an innovation by the department and by the public schools.

The task force produced a document containing its plan for a program leading to certification and to a Masters degree in one calendar year. The plan emphasized the study of teaching through inquiry into *student learning*, *instructional processes*, the *social context* of education, and *leadership* roles. These four emphases were translated into ten goals for the program in order to prepare a teacher who would be knowledgeable about different clients and their needs and proficient in applying instructional strategies.

In the Spring of 1985, the department chair decided to initiate the program and, in general, to follow the recommendations from the task force. After faculty consultation the chair appointed a program director and nine faculty to teach in the program. The program director had not been on the task force. This person had been primarily engaged in conducting research on students' learning but had indicated interest in innovative teacher education programs. Only four of the nine program faculty were from the task force.

The program faculty met to determine specific procedures for admitting persons to the program and for developing the courses which would be taught. In their deliberations, this group developed a set of principles to guide the program which differed to some extent from the goals established by the task force. The program faculty wanted the program to enable scholar teachers to know and use research, to develop a repertoire of teaching strategies, and to make contextual decisions. By dedicating the program to these principles, which reflected their personal commitments, the program faculty began to develop a sense of ownership of the program.

The third of the initial actions by the department chair concerned the involvement of the State Department of Education. In consultation with a representative from the task force and another from the program faculty, the chair submitted a proposal to the Maryland State Department of Education for support of the program. The proposal particularly focused on two needs: to expand the pool of teachers to in-

clude persons with a liberal arts degree and to improve the pedagogical core of teacher preparation programs. These pedagogical improvements were to provide for individualization in the prospective teachers' experiences, to emphasize teaching effectiveness research within a context of problem solving and inquiry, to use new methodologies and technologies (e.g., computer simulations, case studies, microteaching, and action research), and to change field experience opportunities and conceptualizations. Although the financial support requested was not extensive nor essential, the support of the State Department of Education for certification purposes was crucial.

When the program began, the lines of authority were ambiguous. The State Department of Education gave only tentative approval for the program, with the understanding that details would be worked out later; thus, the extent of their authority over the program was undetermined. Also, both the task force and the program faculty claimed ownership of the program. Each was functioning rather autonomously. During the early stages of implementation, the program faculty was busy admitting persons to the program and designing courses. The task force, having completed its work on the proposal for the program, had ceased meeting until the time came for further decisions. Open conflict did not develop until the time came to evaluate and revise the program.

Scholar Teachers

Because the program faculty envisioned a program grounded in theory and research, the persons admitted to the program were referred to as scholar teachers. The criteria for selection were: (a) eligibility for admission to the Graduate School according to established university guidelines (a bachelors degree from an accredited institution, a minimum 3.0 GPA, and a score at or above the 40th percentile on the Miller Analogies Test or the Graduate Record Examination); (b) demonstration of clear commitment to a career in teaching and personal characteristics suitable to teaching; and (c) the applicant's "statement of purpose" as a means to assess thinking and writing skills. The application process also

included a mandatory personal interview conducted by the program faculty.

One constraint upon innovation in curriculum concerns the academic ability of prospective teachers and their commitment to the improvement of teaching. The admissions procedures strongly increased the probability that those entering this program would become scholar teachers committed to improving teaching. The eight scholar teachers admitted for the first year had an average GPA of 3.22, a test score average at the 71st percentile on either the MAT or the GRE, and an average age of 28. Three of them were preparing to teach in elementary schools, and five were preparing for secondary schools in the fields of science, mathematics, English, speech, and theatre. The 26 persons admitted for the second year had slightly higher credentials: an average 3.26 GPA and 73rd percentile test score.

Curriculum

With the admission of the scholar teachers to the program, the faculty turned most of its attention to the design of the curriculum. The three principles developed by the program faculty became the foundation for the design and implementation of the curriculum. The program was to have:

1. a redesigned and integrated teacher education curriculum which was *theory and research based*,
2. learning in a research environment with an emphasis on teacher *problem solving and reflection*, and
3. carefully planned and integrated campus-based and school-based experiences aimed at providing a teaching *repertoire* for the beginning teacher.

The teacher scholars were presented at the time of their entry interview with the expectation that the program would pose rigorous demands due to the number of classes and field experiences that would be required each semester in order to complete the program in one year, and due to the intellectual rigor associated with a graduate program emphasizing theory and research. They were also informed that the program was evolving and experimental and that their ideas and evaluations would be invaluable for the program's successful development.

Courses in the Program. In contrast with most existing teacher preparation

programs, the curriculum was drawn from four fields of inquiry: (a) the research on teaching developed from naturalistic descriptions of effective experienced teachers and from studies of teachers' decision making; (b) research within cognitive psychology which provides new perspectives on the nature of teaching and learning and the ways students approach learning tasks; (c) models of teaching which were developed from theories of learning, motivation, and group dynamics; and (d) research on effective schools. Faculty within this program taught from their own experience and felt an ownership of the content. The intentional emphasis was upon content derived from teaching, developed by teacher educators, and delivered by members of the program faculty. The curriculum began to be truly indigenous to teacher education — a fertile foundation for developing identity in both faculty and scholar teachers.

Courses were specifically designed for this program on these topics: learning principles for teachers, models and processes for teaching, research on teaching, teaching of reading and writing, and classroom diagnosis and measurement. Two courses were adapted for the program (one on the study of teaching and another on curriculum), and institutional modules were provided on group dynamics, leadership, and the social context of schooling. All scholar teachers enrolled in these classes as a cohort group, and they met weekly for the first two semesters in a proseminar. The proseminar included readings, visits to a variety of educational institutions, and discussions about the social context of education. In addition to these collective experiences, they completed special methods classes and content courses in their teaching fields.

Developing a Cohort Group Identity. The first semester of the program (a six-week summer term) was a crucial time in the life of all involved. The scholar teachers coalesced as a cohesive group through sharing their ordeal of being overwhelmed with the curricular demands and the anxiety of producing a scholarly seminar paper. Their development as a group was demonstrated in several events. At the conclusion of their first semester, the scholar teachers planned a social event. The

highlight of their party was the showing of a blooper tape which contained humorous mistakes from their micro-teaching lessons. Their self-concepts and trust in each other were sufficiently strong to display and laugh about their mistakes. The same pattern of coalescing was evident in the second cohort group's experience with the first semester.

Another evidence of growth as a cohort group was shown a few weeks later during the fall semester of the first year. An instructor who was not a member of the program faculty attempted to teach the teacher scholars about group dynamics by leading them through some of the initial stages of group formation. They reported being insulted because they perceived that their group had already matured beyond those beginnings. These illustrations of the experiences of the cohort group demonstrate the importance of this change in the program structure which enables the development of identity.

Experiences for Transferring Theory to Practice. Berliner (1985) has argued that the number one reform in teacher education should be the creation of teaching and pedagogical laboratories where beginners learn to judge their own work. For successful transfer of training, these pedagogical laboratories should include careful presentation of teaching principles, followed by demonstrations, practice with coaching, feedback, and then additional practice (Showers, 1984).

Four features were designed to provide for a better transfer of theory into practice: clinical practice in controlled settings, educational field trips, special practica, and supervised apprenticeships. Clinical practice was provided in a new teacher decision-making and microteaching laboratory. In their first summer session, the scholar teachers presented videotaped lessons according to several models of teaching and received feedback from faculty and peers. They also tutored elementary age students as a way of applying theories of learning.

Most recommendations for innovation in teacher education emphasize particular changes in the field experience either through adding on early field experiences, expanding student

teaching, or developing year-long internships. Our innovation in field experience centered upon the principle of distributed practice. Members of the program faculty were strongly committed to instilling a dialogue between theory and practice and believed that, because initial teaching experiences have such "real-world" attraction, a concerted effort is required to invoke reflection and to avoid unexamined adoption. In order to encourage reflection, the scholar teachers were presented with a diversity of experiential models, they were given theoretical frameworks to use in observing these models, and they were brought out of the field setting on a regular basis in order to reflect upon and discuss their experiences.

During their first semester, the scholar teachers began accumulating a diverse inventory of educational observations. Field trips provided the scholar teachers with firsthand understanding of the context of education. Visits were made to staff development centers, an instructional materials center, the training and development branch of the Government Accounting Office, the Teacher Education Branch of the Maryland State Department of Education, a model private school, a model special education center, and a regional conference on beginning teacher programs. The field trips were linked to readings and discussions, particularly in the proseminar.

Beginning in the second semester, the scholar teachers increased their participation in the field settings. All scholar teachers attended portions of the first week of school in selected elementary and secondary schools in order to find instances of their readings about effective beginnings of the school year. They piloted a new classroom management observational system being developed by Sanford and Development Center for Teacher Education (e.g., Evertson, Emmer, Sanford, and Clements, 1983). They also conducted focused observations of time use as an application of their reading and discussion of the research on academic learning time and other studies of teaching effectiveness.

Their supervised apprenticeship came in the third of the four semesters

in the program. This experience was also designed to provide distributed practice rather than the typical massed practice. Plans were made to pull the beginning teachers out of the school settings after four weeks for a two-week period of reflection, and then to repeat this cycle again. Although strong objections were raised by certain faculty, by the cooperating teachers in the public schools, and by the scholar teachers, a modification of this "distributed" approach was implemented. Throughout their experiences in the schools, the scholar teachers were expected proactively to monitor and construct the learning environment; they carefully avoided "unthinking" conformity to the schools' practices. By the end of that semester, most of the scholar teachers had expressed appreciation for the objectivity they gained about what was happening in the schools as a result of the distributed practice and the participation in multiple classrooms and schools.

Problem-Solving Activities. The curriculum of this program is built on the premise that research can guide practice but should not be translated into a set of recipes and formulas for teachers to follow blindly. Acquisition of a knowledge base of research on teaching and learning is necessary but not sufficient to ensure that prospective teachers become skilled and reflective practitioners. Knowledge produced by educational research, by its very nature, is applicable to the way things work in general and not to specific cases. The problems faced by teachers in real classrooms are characterized by uniqueness, contextual complexity, instability, and value conflicts.

The program faculty hoped to develop in the scholar teachers the attitude that learning to teach is a life-long process, and that teachers themselves can help discover their own best practice through reflection and critical inquiry, guided by a knowledge base derived from research on teaching and learning. Procedures and strategies in the program designed to promote reflective teaching and decision making included a proseminar, self-help laboratories, keeping a journal, a seminar paper, and an action research project.

Scholar teachers regularly attended a proseminar where they reflected

about the week's activities and submitted all academic and field-based experiences to critical review and analysis. To promote this reflection, the scholar teachers were asked to regularly record and analyze their experiences in a journal. Another regular feature of the weekly proseminar was time devoted to self-help labs where scholar teachers applied problem solving strategies to situations they had observed in field settings or to components of the program they considered unsatisfactory. Their recommendations resulted in several adjustments to the program. For example, at the beginning of an internship they felt a need to spend more time in the school setting. Program faculty agreed to rearrange the university course schedules to allow complete engagement in the school setting for two weeks.

Two major research projects were required for all scholar teachers. The first was a library-based project requiring them to gain mastery of a particular field of knowledge by doing an extensive search of the literature and a careful synthesis of the knowledge base on a particular topic drawn from their studies in learning and teaching. For their second project, the scholar teachers identified a researchable problem and conducted a classroom-based action research project.

The Process of Innovation and Evaluation

While the program was initiated with specific plans for curricular content and for formal evaluations, the norm quickly emerged that evaluation would be informal and continuous. The program faculty respected the scholar teachers' capacity to critique and contribute to their experience; the scholar teachers regularly gave thoughtful recommendations. In the proseminar, the scholar teachers were consulted weekly in order to make collaborative decisions regarding their field visits and other curricular plans. Meetings were called when collective decision making was needed rather than according to a prearranged calendar.

The scholar teachers' comments on their formal evaluations for the first half of the program confirmed the issues that had already been suggested through the informal process. They ex-

pressed their support for the first seminar paper. Although stressful at the time, the process of producing the scholarly paper was viewed with pride. They valued the theoretical orientation to the curriculum and resented the few times that teaching strategies seemed to be presented as "a bag of tricks." The proseminar was considered valuable, particularly the help labs and the diversity in field experiences. They felt that their understanding of group dynamics had developed through experience, and they did not need additional classroom instruction about it.

In the beginning of the spring semester of the first year, the program faculty spent a day and a half reviewing the suggestions of the scholar teachers, evaluating the program, and making plans for the second cycle. The meeting followed a problem solving procedure of reviewing issues, suggesting resolutions and modifications, using the program goals as criteria, and then making specific plans for action. This process enabled the faculty to strengthen their commitment to the goals of the program as each goal was defended and evaluated. The infusion of theory and research into the program was found to be most successful; a repertoire of teaching models was viewed as insufficiently developed; and problem solving and reflection seemed generally successful but could be promoted through more modeling in the field settings. Action steps included a number of modifications for the program as well as a commitment to derive a conceptual approach to the development of a repertoire. Additional readings in philosophy of education were incorporated into classes, and a module on the use of the computer was designed and included in the second year.

As the program faculty planned for a new cycle of the program, two issues came to the surface. The program faculty and the task force were no longer working on discrete tasks. Who had authority to make decisions? After working extensively in the program for two semesters, the program faculty felt sufficient ownership that they were willing to struggle for more autonomy. Following extensive discussions, the task force agreed to assume an advisory posture so that the program faculty could efficiently direct the program

and extend their commitment to it.

The second issue that surfaced at this time was termed "old baggage." The old baggage that the program kept trying to give away included a college-wide requirement and a departmental requirement; in its existing form, each was considered by the program faculty to be inappropriate for the scholar teachers. For example, the college requirement was a particular class in statistics. While the program faculty supported the inclusion of a rigorous statistics course, they argued for more adaptations thereby making the applications more relevant to research used and conducted by teachers. These requirements were finally resolved through a long series of compromises, taking a heavy toll on the spirit of innovation.

Before drawing from this program any general conclusions about innovation in teacher education, we should place the current status of the program in perspective. Seven of the eight persons in the first group were hired in school systems of their choice, and the eighth decided to continue his education in order to be certified in an additional teaching field. The second group has 25 persons who appear to be excellent prospective teachers. Plans for a third cycle continue with more qualified applicants than positions. While the program has been accepted by the Department, the College of Education, and the State Department of Education, it is only in its second year. Approvals involving the formal university structure require more time and effort. While the innovation still faces many tests, the successes of the first year-and-a-half merit tentative conclusions.

Findings About Innovation

Although we have tried to separate the themes of identity, passivity, curriculum, and control for purposes of analysis and discussion, their fusion in the fabric of the program is certainly evident. For example, as the scholar teachers came to identify with the program and with the profession of teaching, they simultaneously performed as problem solvers, taking an active role in shaping the program and their own professional development. Acknowledging these interrelationships, we

will again address them separately to present what we feel are insights into successful innovation.

Identity

Prospective teachers and university faculty can identify with a program in teacher education and with the profession of teaching. Contributing factors to this process include having a content that belongs to teaching and is central to the program, developing a sense of cohesion through experiences as a cohort group for both faculty and prospective teachers, and having the responsibility and authority to make significant decisions about the functioning of the program.

Faculty whose personal research focuses on teaching and learning can identify professionally with teacher education as they work collectively with colleagues and capable teacher scholars. Contributing to this identity is the opportunity to conduct their research within the program and to include their research in the content of the curriculum. Thus the professional identity strengthens the identity with the program. For both scholar teachers and program faculty, perception of the program as rigorous and demanding contradicts "the long held view that there is not much to teaching," a primary criticism raised in *Educating a Profession* (Howsam et al., 1976, p. 11).

Activity (Non-Passivity)

Prospective teachers and university faculty can assume active roles within programs of teacher education. For example, faculty can design a curriculum with structures such as proseminars in which scholar teachers solve problems rather than passively receive information. Faculty can model problem solving by involving the scholar teachers in their research projects and by externalizing their own decision processes about their research and teaching. Both groups can be involved in the continuous evaluation and modification of the program. Contrary to the criticism of traditional programs, this program emphasized "investigation, reflection, and solving novel problems," a vital missing link according to the Holmes group (1986, p. 62).

Control

We found no easy answers for the classic struggle between change agents

and power brokers. Scannell (1986) documented how resistance increases over time as new programs move from "good ideas" to specific infringements on vested powers. Our program faculty, acting as change agents, demanded enough autonomy to explore their ideas apart from the interests of those representing the existing programs. Resistance has continued and even increased particularly through other faculty demanding additional requirements representing their special interests. The support of the department chair and a favorable review by the State Department of Education led the task force and other faculty to general acceptance of the program faculty's autonomy.

The key to diffusing innovative ideas into existing programs may depend on the activity of those persons having membership in both groups, change agents and power brokers. Can the persons with overlapping involvement (in our case those four persons having membership in both the program faculty and the task force) be given sufficient support and responsibility to make this happen? In our case, these persons have enlisted general support for the innovative program; they continue to resolve specific issues.

The theme of control concerns scholar teachers as well as the faculty. Teacher autonomy was one of three themes of reform drawn from the background papers from the National Commission for Excellence in Teacher Education. While autonomy seems needed for successful reform, Lasley and Rogus (1986) raise the question about how long teachers valuing autonomy will persist in a school culture which is characterized by a "go-along" attitude. Certainly we need to follow up our graduates to see if they become demoralized or if they are able either to find supportive schools or to help shape schools into supportive climates for persons who want to participate in making important decisions.

Curriculum

Our experiences with the program have convinced us of the dynamic relationship between curricular reform and the preceding three themes. For innovation to be designed and implemented in the face of inevitable pressures to maintain the status quo, pro-

gram personnel must have a strong sense of identity with and commitment to the program, they must take a continued proactive stance in demanding the right to change, and they must exercise enough control to influence change.

We believe this innovation is successful because the curriculum has: (a) substantive, conceptual goals that are identified at the beginning and that represent convictions of the program faculty; (b) a content that can be owned by teacher education that reflects "legitimate professional knowledge, skills, and an ethos of responsibility" (Holmes Group, 1986, p. 55); and (c) continuous and serious evaluation of the curriculum. With the program just described, the nature of the goals demanded such evaluation, and the evaluation and modifications strengthened the commitment of the faculty and the scholar teachers to the goals.

Through our experiences with this program we say along with the writers of the Holmes report (1986):

But we have also found ourselves willing to argue for radical improvements that most of us would have dismissed as impractical just a few years ago. We have decided that we must work for the changes that we believe to be right, rather than those that we know can succeed. Much is at stake, for American students' performance will not improve much if the quality of teaching is not much improved. And teaching will not improve much without drastic improvements in teacher education. (p. 1)

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